

V3 empirical study · Instruction

Seven models · four underlyings · four volatility regimes · 1.5-year fixed calibration

What this report is

- Computational source of truth: the companion Jupyter notebook. This PDF is written from the same payload; every table number is identical.
- Question: which of 7 American-call pricing models tracks listed market prices most closely, and does the ranking change across companies and volatility regimes?
- In every table the BEST row is the lowest percentage RMSE. That row is bold and shaded green, and the Mark column says BEST.

Experimental design (held fixed for every model)

- Models: GBM, Modified GBM (Markov direction), GARCH(1,1), Heston (no jumps), Merton jump-diffusion, GARCH-Merton, Heston-Merton (Bates).
- Modified GBM (Markov direction + split-normal magnitudes) is specified in `V3_modified_gbm_model.pdf` in this folder.
- Companies: SPY (primary), AAPL, MSFT, AMZN. Each name is calibrated and priced separately.
- Regimes (V3 one-year evaluation windows): Crisis 2008-08-01→2009-07-31; Normal 2014-01-01→2014-12-31; Late-cycle 2018-10-01→2019-09-30; COVID 2019-09-01→2020-08-31.
- The late-cycle and COVID files share September 2019 (V3 notebook dates). All other regime days are disjoint. Contracts are built independently in each window.
- Calibration window: 1.5-year ending at the first session of each 1-year evaluation window. Rolling: none (one estimate, held for the whole window). Parameters estimated at t_0 are used only after t_0 .
- If history is shorter than the requested lookback, the estimator uses all available observations up to the update date (equity starts 2003-12-01; listed-option panels start 2008-01).

V3 empirical study · Instruction (continued)

Seven models · four underlyings · four volatility regimes · 1.5-year fixed calibration

Shared market sample and filters (not model-specific)

- For each company $\times$ regime, one listed-call sample is drawn once and reused by all seven models in this report: nearest-ATM call on each Monday (next session if Monday is closed), DTE 7–60, listed expiry, as-of that date.
- Estimation filters, applied in order to every quote used for Heston/Heston–Merton calibration and to the LSM comparison set: (1) calls only, finite S , K , C ; (2) no-arbitrage $C \geq \max(0, S-K)$; (3) $7 \leq \text{DTE} \leq 60$; (4) $|S/K - 1| \leq 10\%$; (5) premium ≥ 0.05 , valid bid-ask with relative spread $\leq 50\%$ when those columns exist, volume ≥ 1 when present.
- Return-based models (GBM, Merton, GARCH, GARCH–Merton) never see option quotes in §4; they still price the same filtered LSM contracts in evaluation.

Model-correctness adjustments (V3)

- Heston / Heston–Merton: $(\kappa, \theta, \xi, \rho, v_0)$ are option-implied Fourier NLS (Albrecher little-trap), not Method A realized-variance moments. μ is the P-measure lookback mean of stock returns; LSM replaces $\mu \rightarrow r$.
- Euler ordering: the stock increment over $[t, t+\Delta t]$ uses the current variance v_t ; $v_{t+\Delta t}$ is updated afterwards. No look-ahead in the variance.
- LSM: Longstaff–Schwartz American call on the full path cloud ($n_{\text{paths}} = 10000$, seed 42, $n_{\text{steps}} = \text{DTE}$, $\Delta t = 1/252$). Paths are not averaged before stopping.

Metrics

- Primary: $\text{RMSE\%} = 100 \times \sqrt{\text{mean}(((C_{\text{model}} - C_{\text{mkt}}) / C_{\text{mkt}})^2)}$. Ranking uses this number only.
- Secondary: $\text{MAE} = \text{mean} |C_{\text{model}} - C_{\text{mkt}}|$ (dollars). $\text{Bias \$} = \text{mean}(C_{\text{model}} - C_{\text{mkt}})$ (dollars). $\text{Bias\%} = 100 \times \text{mean}((C_{\text{model}} - C_{\text{mkt}})/C_{\text{mkt}})$. Positive bias = model expensive vs market. Early-exercise fraction = mean share of LSM paths that stop before expiry.

Page 3 · Filters, sample, and no-look-ahead

Shared evaluation sample · n listed calls (identical for all seven models in this report)

Regime	Window	SPY n	AAPL n	MSFT n	AMZN n
2008-2009 Crisis	2008-08-01 → 2009-07-31	50	50	50	50
2013-2014 Normal	2014-01-01 → 2014-12-31	50	50	50	50
2018-2019 Late-cycle	2018-10-01 → 2019-09-30	50	51	50	51
2019-2020 COVID	2019-09-01 → 2020-08-31	50	50	50	50

Filter funnel · last-step n in the 1.5-year lookback ending at each regime (processed call panel)

Ticker	Regime	Input*	No-arb	DTE 7-60	S/K-1 ≤10%	Liquidity
SPY	2008-2009	3359	3311	3311	3203	3203
SPY	2013-2014	12815	12548	12548	12405	12405
SPY	2018-2019	35222	35104	35104	34809	34809
SPY	2019-2020	42100	41961	41961	41379	41379
AAPL	2008-2009	685	685	685	652	652
AAPL	2013-2014	8925	8875	8875	8633	8633
AAPL	2018-2019	6309	6297	6297	6084	6084
AAPL	2019-2020	8010	7992	7992	7713	7713
MSFT	2008-2009	566	566	566	538	538
MSFT	2013-2014	3299	3237	3237	3175	3175
MSFT	2018-2019	7814	7752	7752	7583	7583
MSFT	2019-2020	6892	6852	6852	6655	6655
AMZN	2008-2009	465	465	465	442	442
AMZN	2013-2014	5715	5692	5692	5598	5598
AMZN	2018-2019	28809	28808	28808	28346	28346
AMZN	2019-2020	26766	26760	26760	26323	26323

*Input is the processed call panel restricted to (regime end – 1.5-year, regime end]. Processed files already drop many raw quotes; the funnel above is the additional V3 research filter.

No look-ahead. Monthly parameters at update date t use only equity returns and listed quotes with trading_date $\leq t$. LSM prices the Monday contract using the latest calibration row with date $\leq$ that Monday. Quotes are as-of the grid timestamp (same session, else the prior session).

SPY 2008–2009 uses bid-ask mid when the source mark is \$0.01 (a GitHub field error, not missing trades). After that repair the processed panel has weekly coverage, so n matches AAPL/MSFT (~50). All seven models in this report still share that exact sample.

Heston / Heston-Merton calibration further subsamples the filtered lookback surface to ≤ 24 contracts (maturity $\times$ moneyness strata). GBM / Merton / GARCH / GARCH-Merton estimate from lookback log returns only (jumps: 3σ residual threshold where applicable).

Page 4 · Table block · SPY · seven models × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, rolling=none, n_paths=10000, seed=42.

Table SPY-1 SPY · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	39.85	1.5118	-1.3570	-29.27	25.5%	5
Modified GBM	43.13	1.6255	-1.5635	-35.30	26.1%	7
GARCH	35.72	1.3442	-0.8523	-14.47	21.6%	2
Heston	37.99	1.4513	-1.1865	-24.31	29.0%	4
Merton	39.86	1.5142	-1.3598	-29.38	24.3%	6
GARCH-Merton	35.69	1.3398	-0.8186	-13.54	20.8%	1
Heston-Merton	37.76	1.4404	-1.1711	-23.83	27.2%	3

Table SPY-2 SPY · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	40.10	0.7145	+0.5358	+27.51	26.2%	2
Modified GBM	41.67	0.7464	+0.5698	+29.03	26.1%	4
GARCH	40.43	0.7184	+0.5469	+27.93	23.4%	3
Heston	80.89	1.6010	+1.5789	+70.66	25.2%	6
Merton	39.30	0.7016	+0.5143	+26.61	24.2%	1
GARCH-Merton	45.88	0.8274	+0.6986	+34.18	22.4%	5
Heston-Merton	80.98	1.6058	+1.5835	+70.80	25.2%	7

Table SPY-3 SPY · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	23.62	0.8840	-0.1277	+1.81	20.8%	5
Modified GBM	23.03	0.8902	-0.2283	-0.59	22.2%	2
GARCH	25.80	1.0497	-0.8867	-16.54	17.2%	6
Heston	26.62	0.9285	+0.1483	+7.58	19.5%	7
Merton	23.41	0.8884	-0.1839	+0.37	14.3%	4
GARCH-Merton	22.86	0.9032	-0.4350	-5.93	16.7%	1
Heston-Merton	23.17	0.9016	-0.3351	-3.49	11.7%	3

Table SPY-4 SPY · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	57.52	2.9208	-0.5111	+18.64	24.3%	2
Modified GBM	59.99	2.9847	-0.3345	+21.88	23.9%	3
GARCH	63.77	3.0315	-0.0942	+26.41	21.2%	4
Heston	80.30	3.3486	+0.8238	+44.02	30.1%	6
Merton	56.08	2.9267	-0.6252	+16.46	20.8%	1
GARCH-Merton	70.25	3.1595	+0.2869	+33.51	20.3%	5
Heston-Merton	86.96	3.5301	+1.2470	+51.47	23.5%	7

RMSE% is $100 \times \sqrt{\text{mean}(((\text{model} - \text{market}) / \text{market})^2)}$. MAE and Bias \$ are in option-premium dollars. Bias% is $100 \times \text{mean}((\text{model} - \text{market}) / \text{market})$. Early-ex. frac. is the mean LSM share of paths e

Page 5 · Table block · AAPL · seven models × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, rolling=none, n_paths=10000, seed=42.

Table AAPL-1 AAPL · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	34.70	1.7325	-0.2012	+2.49	23.4%	5
Modified GBM	32.72	1.6706	-0.6554	-4.84	24.4%	1
GARCH	40.04	1.9147	+0.4172	+12.21	17.2%	6
Heston	34.03	1.7307	-0.2640	+1.23	22.3%	2
Merton	34.26	1.7225	-0.2873	+0.95	21.0%	4
GARCH-Merton	43.53	2.0289	+0.7015	+16.92	16.7%	7
Heston-Merton	34.10	1.7166	-0.2799	+1.17	19.5%	3

Table AAPL-2 AAPL · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	57.70	3.8194	+3.8194	+51.38	24.4%	4
Modified GBM	51.10	3.3484	+3.3484	+44.85	25.7%	2
GARCH	45.96	2.9301	+2.9021	+39.06	20.4%	1
Heston	58.84	3.9883	+3.9883	+52.30	19.8%	5
Merton	52.75	3.5099	+3.5099	+46.69	20.4%	3
GARCH-Merton	67.13	4.6054	+4.6054	+61.25	17.2%	7
Heston-Merton	58.99	3.9831	+3.9831	+52.19	17.1%	6

Table AAPL-3 AAPL · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 51 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	16.77	0.8809	-0.5584	-7.11	23.8%	2
Modified GBM	16.88	0.8919	-0.5478	-6.91	24.5%	3
GARCH	16.17	0.8273	-0.3825	-4.32	19.9%	1
Heston	31.34	1.5900	+1.3922	+23.98	19.8%	7
Merton	17.13	0.9184	-0.6295	-8.25	21.1%	4
GARCH-Merton	17.34	0.8370	+0.0909	+3.31	19.3%	5
Heston-Merton	21.47	1.0270	+0.5515	+10.59	22.7%	6

Table AAPL-4 AAPL · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	41.39	3.1111	+0.6446	+18.83	24.0%	2
Modified GBM	43.50	3.2399	+0.9173	+21.54	24.2%	3
GARCH	54.11	3.9481	+2.2445	+34.15	20.1%	6
Heston	45.09	3.3073	+1.1657	+23.85	26.1%	4
Merton	39.55	3.0542	+0.3844	+16.28	20.1%	1
GARCH-Merton	68.46	5.0306	+3.8961	+50.08	18.7%	7
Heston-Merton	45.11	3.3287	+1.1442	+23.69	20.6%	5

Page 6 · Table block · MSFT · seven models × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, rolling=none, n_paths=10000, seed=42.

Table MSFT-1 MSFT · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	29.17	0.3120	-0.2729	-19.88	25.5%	4
Modified GBM	28.96	0.3119	-0.2681	-19.42	26.3%	3
GARCH	29.41	0.2879	+0.0218	+8.10	19.6%	5
Heston	27.04	0.3008	-0.2200	-14.34	27.4%	1
Merton	30.20	0.3213	-0.2903	-21.66	23.7%	6
GARCH-Merton	32.88	0.2963	+0.0849	+14.04	20.0%	7
Heston-Merton	27.09	0.3016	-0.2193	-14.22	23.5%	2

Table MSFT-2 MSFT · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	62.49	0.4284	+0.4262	+52.88	24.7%	4
Modified GBM	60.58	0.4138	+0.4100	+51.07	26.0%	2
GARCH	73.22	0.5232	+0.5232	+63.91	21.0%	6
Heston	62.26	0.4275	+0.4254	+52.75	24.5%	3
Merton	55.05	0.3688	+0.3615	+45.40	17.6%	1
GARCH-Merton	96.85	0.7307	+0.7307	+87.77	17.1%	7
Heston-Merton	62.79	0.4308	+0.4271	+53.09	18.6%	5

Table MSFT-3 MSFT · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	21.68	0.5626	+0.0129	+3.25	23.0%	4
Modified GBM	19.94	0.5524	-0.2155	-3.65	25.4%	1
GARCH	20.87	0.5485	-0.0844	+0.24	18.6%	2
Heston	28.71	0.7076	+0.4085	+15.07	22.6%	6
Merton	21.14	0.5548	-0.0675	+0.85	17.2%	3
GARCH-Merton	25.36	0.6383	+0.2795	+11.22	16.3%	5
Heston-Merton	30.54	0.7557	+0.4887	+17.50	20.4%	7

Table MSFT-4 MSFT · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	45.98	1.7242	+0.0264	+16.93	24.8%	5
Modified GBM	41.04	1.7043	-0.3996	+8.84	25.5%	1
GARCH	45.83	1.7300	+0.0064	+16.56	20.1%	4
Heston	47.96	1.7733	+0.1593	+19.55	25.7%	6
Merton	45.04	1.7309	-0.0492	+15.51	21.8%	2
GARCH-Merton	53.39	1.8833	+0.4719	+25.84	18.4%	7
Heston-Merton	45.23	1.7365	-0.0483	+15.57	21.2%	3

Page 7 · Table block · AMZN · seven models × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, rolling=none, n_paths=10000, seed=42.

Table AMZN-1 AMZN · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	32.38	1.2336	-0.0334	+3.64	22.9%	4
Modified GBM	29.85	1.2017	-0.4075	-4.77	25.2%	3
GARCH	38.93	1.3651	+0.4984	+15.15	15.9%	6
Heston	29.77	1.2161	-0.3467	-3.72	24.3%	2
Merton	32.42	1.2247	-0.0467	+3.32	17.7%	5
GARCH-Merton	53.16	1.7374	+1.2478	+32.42	15.6%	7
Heston-Merton	28.96	1.1935	-0.5002	-7.24	14.4%	1

Table AMZN-2 AMZN · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	23.09	2.4183	+0.0489	+5.54	24.8%	3
Modified GBM	22.52	2.3818	-0.0168	+4.90	25.5%	2
GARCH	25.31	2.4409	+0.5868	+10.13	19.9%	4
Heston	59.01	5.2473	+5.1382	+49.55	19.3%	7
Merton	22.03	2.3804	-0.1930	+3.35	19.9%	1
GARCH-Merton	35.61	2.9530	+2.1822	+24.10	18.2%	5
Heston-Merton	44.80	3.7149	+3.3705	+34.36	21.4%	6

Table AMZN-3 AMZN · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 51 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	22.97	12.4353	-5.5218	-4.77	22.9%	1
Modified GBM	23.72	12.4787	-3.3465	-1.02	21.5%	4
GARCH	23.59	12.6060	-2.9523	-0.74	18.5%	2
Heston	53.74	24.7431	+22.4495	+40.39	19.5%	7
Merton	23.70	12.7887	-8.1593	-9.42	15.5%	3
GARCH-Merton	28.66	13.9181	+3.7835	+10.56	14.3%	5
Heston-Merton	48.52	21.8398	+18.6520	+34.17	18.0%	6

Table AMZN-4 AMZN · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	56.86	26.5091	+13.6258	+35.54	23.7%	5
Modified GBM	60.74	27.6000	+16.4384	+39.62	22.9%	7
GARCH	41.58	23.0490	+0.2550	+16.22	21.4%	2
Heston	39.20	23.2576	-2.6623	+12.80	28.2%	1
Merton	55.02	26.1281	+12.4842	+33.71	21.1%	4
GARCH-Merton	44.57	23.4896	+3.5116	+20.71	21.1%	3
Heston-Merton	58.56	27.2428	+15.5745	+37.07	19.1%	6

Page 7 · Summary tables · RMSE% across companies and regimes

Table S1 · Overall ranking (mean RMSE% over 16 cells)

Rank	Model	Mean RMSE%	Median RMSE%	# of 16 cells best
1	Merton	36.68	36.78	5
2	Modified GBM	37.46	36.88	3
3	GBM	37.89	37.28	1
4	GARCH	38.80	39.48	2
5	Heston-Merton	45.94	44.95	1
6	GARCH-Merton	46.35	44.05	2
7	Heston	46.43	42.15	2

Table S2 · Wins by company (lowest RMSE% in that company×regime)

Model	SPY wins	AAPL wins	MSFT wins	AMZN wins	Total
GBM	0	0	0	1	1
Modified GBM	0	1	2	0	3
GARCH	0	2	0	0	2
Heston	0	0	1	1	2
Merton	2	1	1	1	5
GARCH-Merton	2	0	0	0	2
Heston-Merton	0	0	0	1	1

Table S3 · Mean RMSE% by regime (equal-weight mean of SPY / AAPL / MSFT / AMZN)

Model	2008-2009 Crisis	2013-2014 Normal	2018-2019 Late-cycle	2019-2020 COVID	Mean of 4
GBM	34.03	45.85	21.26	50.44	37.89
Modified GBM	33.66	43.97	20.89	51.32	37.46
GARCH	36.02	46.23	21.61	51.32	38.80
Heston	32.21	65.25	35.10	53.14	46.43
Merton	34.18	42.28	21.35	48.92	36.68
GARCH-Merton	41.32	61.37	23.55	59.17	46.35
Heston-Merton	31.98	61.89	30.93	58.97	45.94

Green row in S1 / S3 = lowest mean RMSE%. Green row in S2 = most company×regime wins. A model can win the most cells without having the lowest average error (and vice versa).

Table S4.1 · SPY

Model	Mean RMSE%	Median	# regimes best
GBM	40.27	39.98	0
Modified GBM	41.95	42.40	0
GARCH	41.43	38.07	0
Heston	56.45	59.15	0
Merton	39.66	39.58	2
GARCH-Merton	43.67	40.78	2
Heston-Merton	57.22	59.37	0

Table S4.2 · AAPL

Model	Mean RMSE%	Median	# regimes best
GBM	37.64	38.04	0
Modified GBM	36.05	38.11	1
GARCH	39.07	43.00	2
Heston	42.33	39.56	0
Merton	35.92	36.90	1
GARCH-Merton	49.12	55.33	0
Heston-Merton	39.92	39.61	0

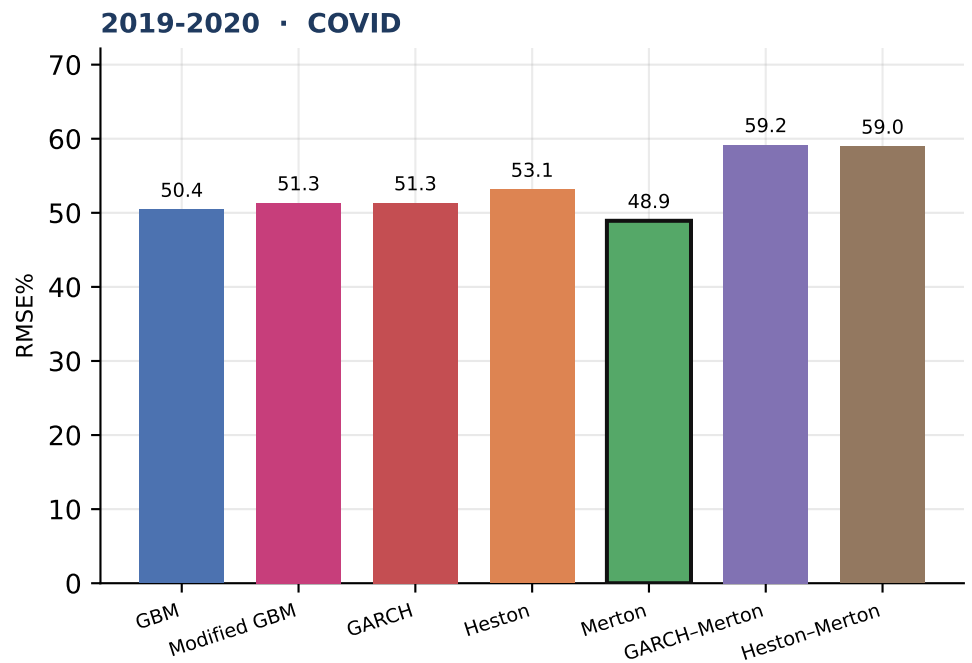
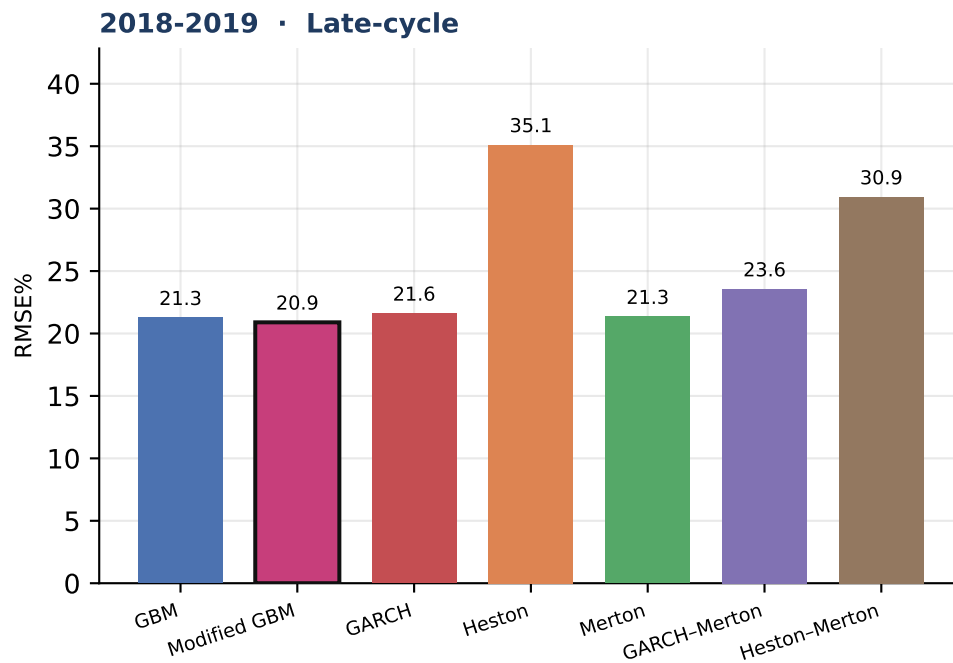
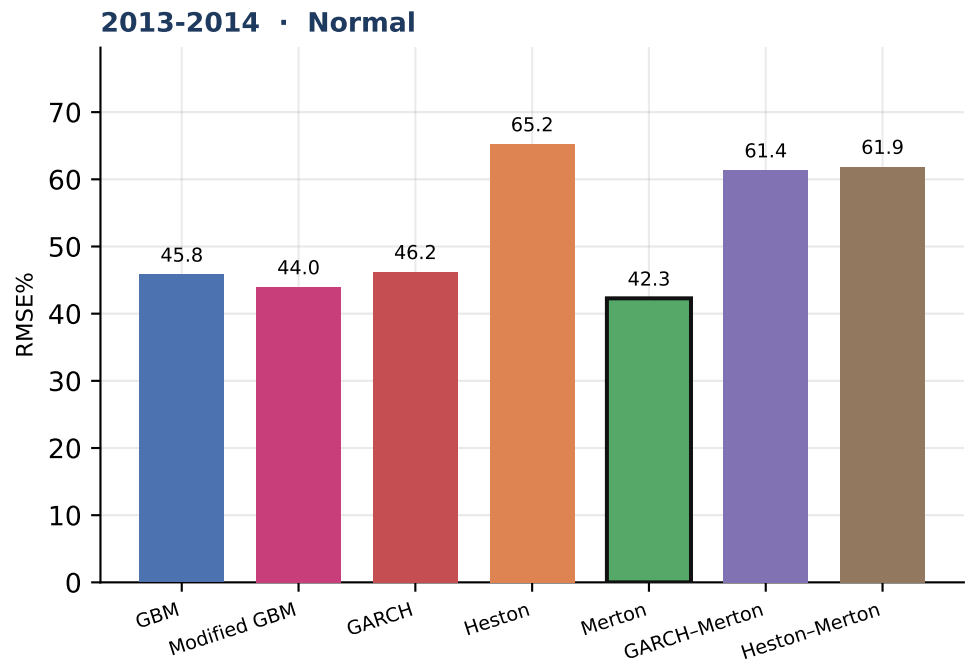
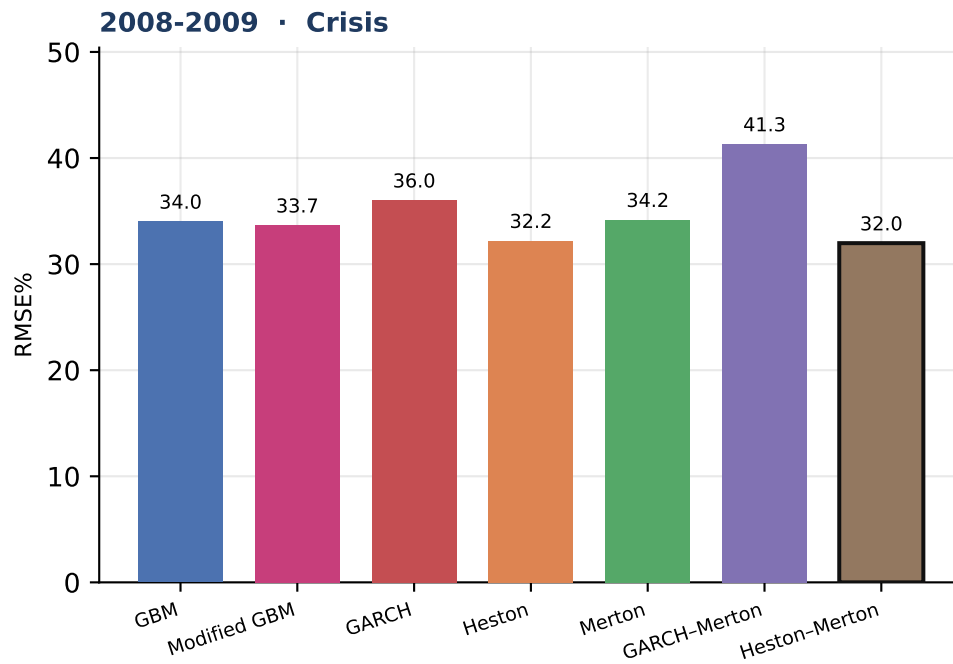
Table S4.3 · MSFT

Model	Mean RMSE%	Median	# regimes best
GBM	39.83	37.57	0
Modified GBM	37.63	35.00	2
GARCH	42.33	37.62	0
Heston	41.49	38.33	1
Merton	37.86	37.62	1
GARCH-Merton	52.12	43.14	0
Heston-Merton	41.41	37.89	0

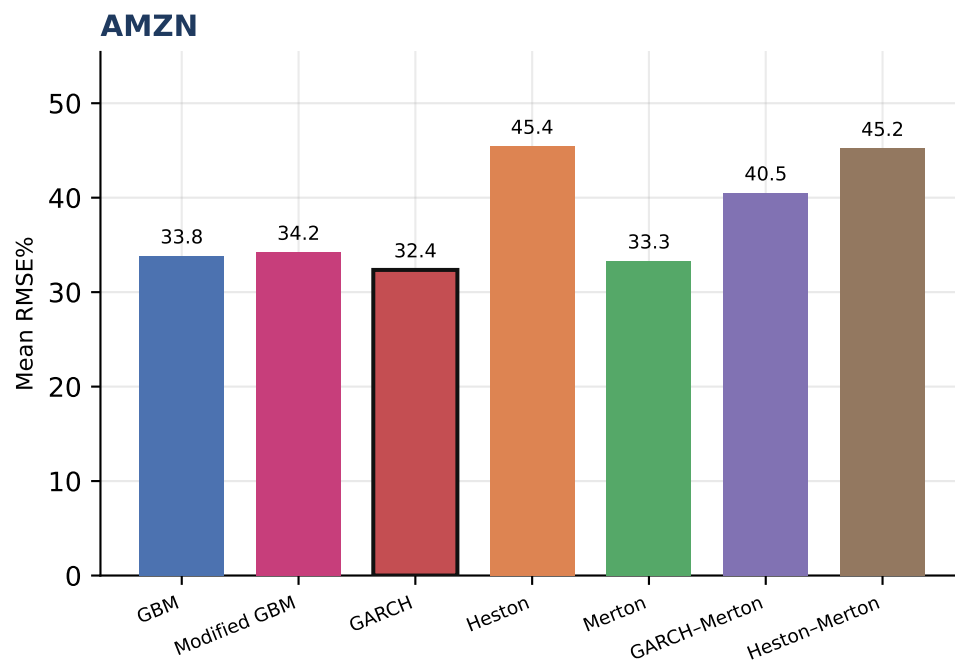
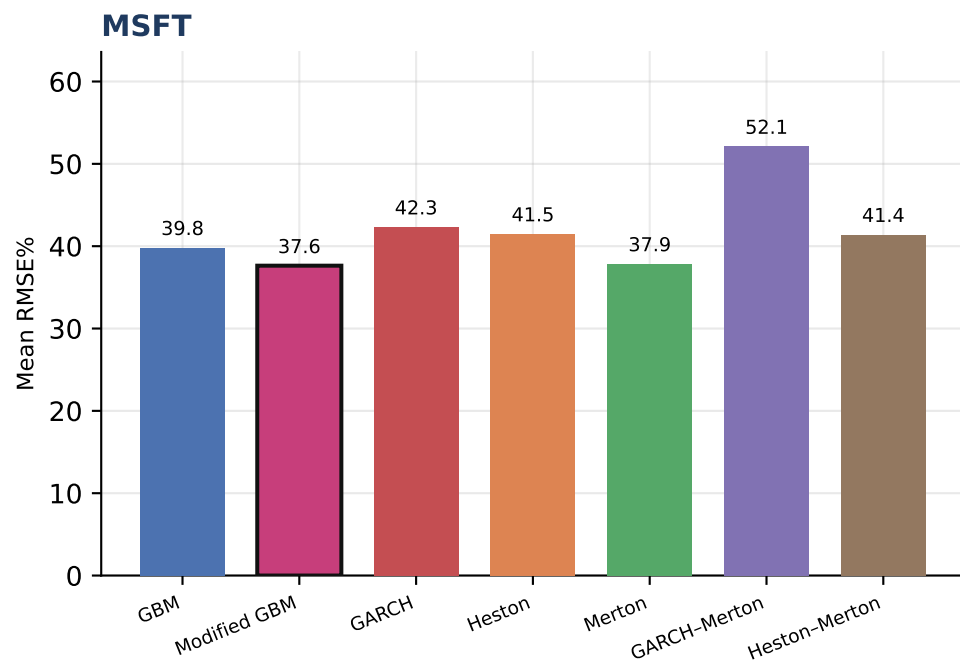
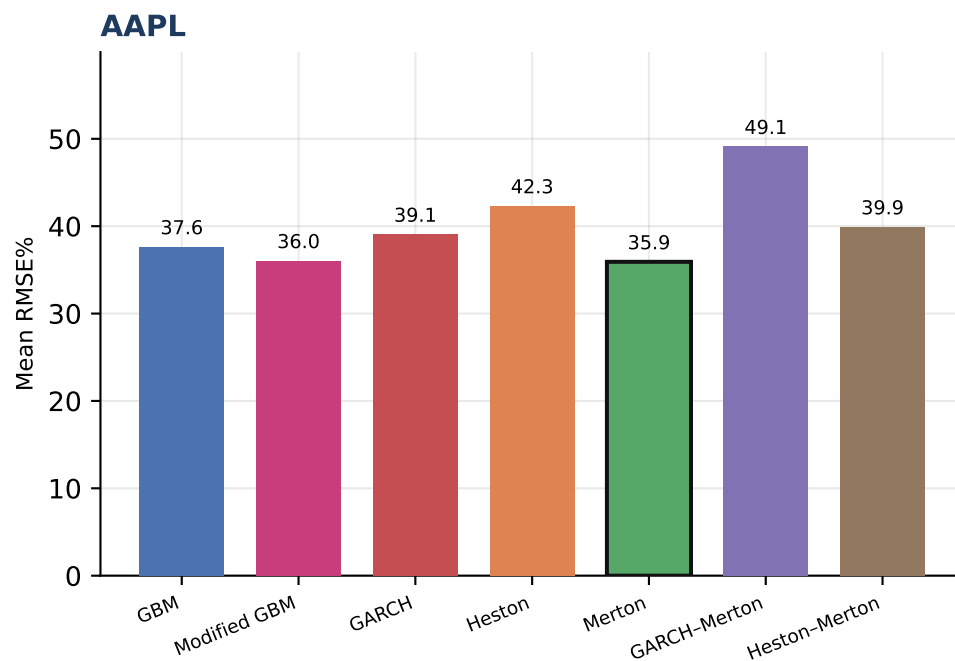
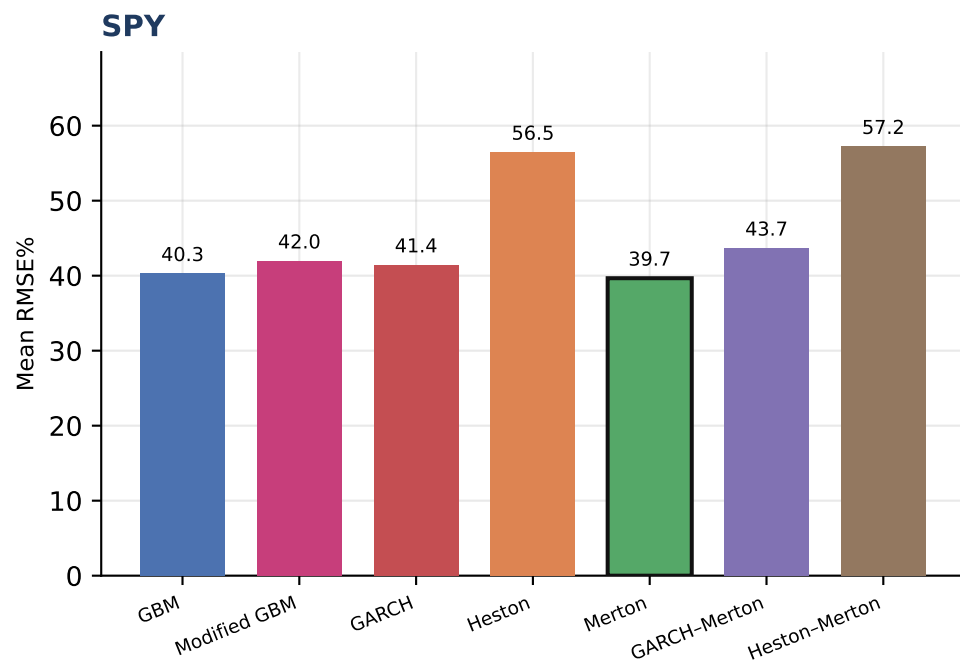
Table S4.4 · AMZN

Model	Mean RMSE%	Median	# regimes best
GBM	33.82	27.73	1
Modified GBM	34.21	26.79	0
GARCH	32.35	32.12	0
Heston	45.43	46.47	1
Merton	33.29	28.06	1
GARCH-Merton	40.50	40.09	0
Heston-Merton	45.21	46.66	1

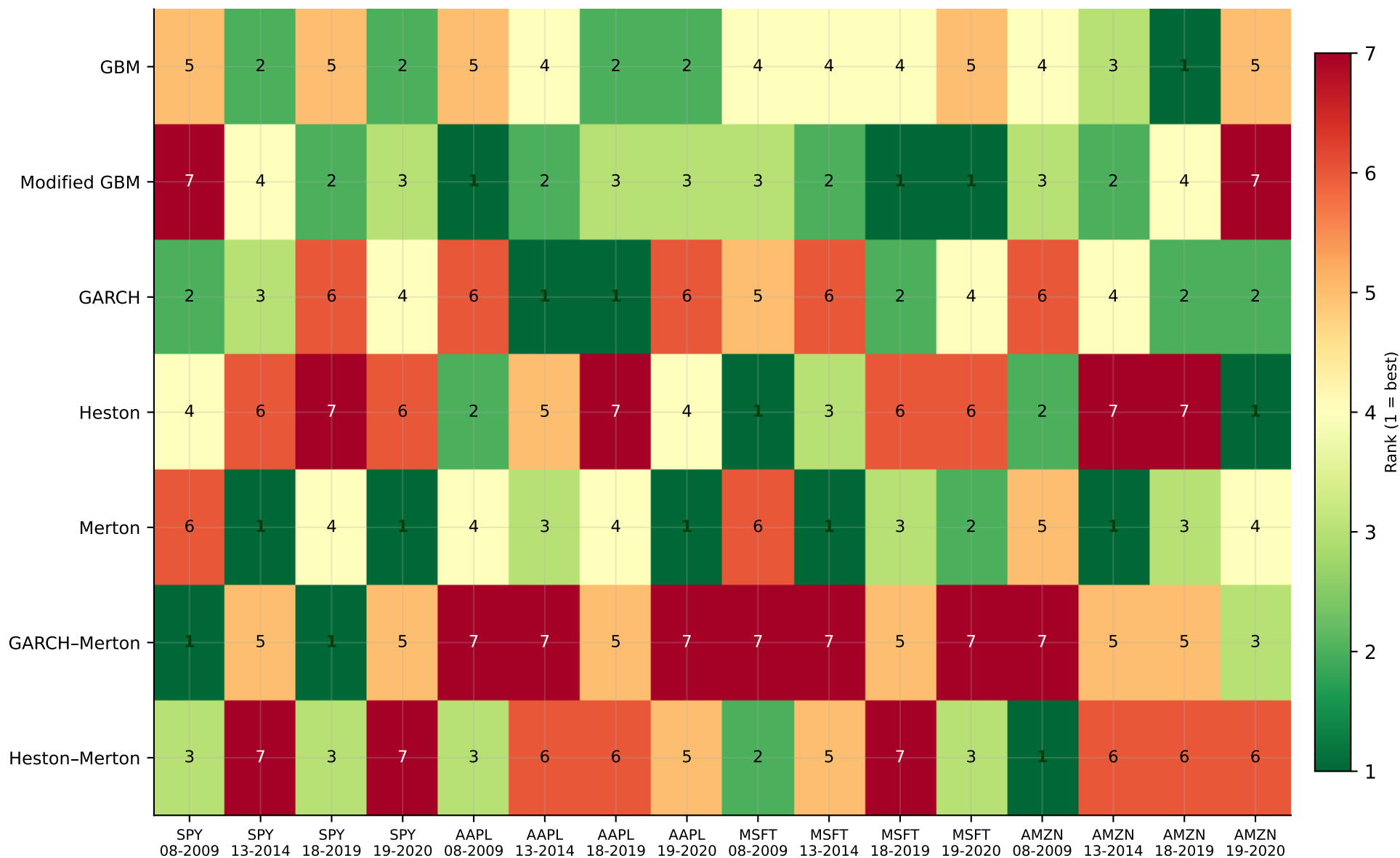
Each panel is one company. Green row = lowest mean RMSE% across the four V3 regimes for that name.



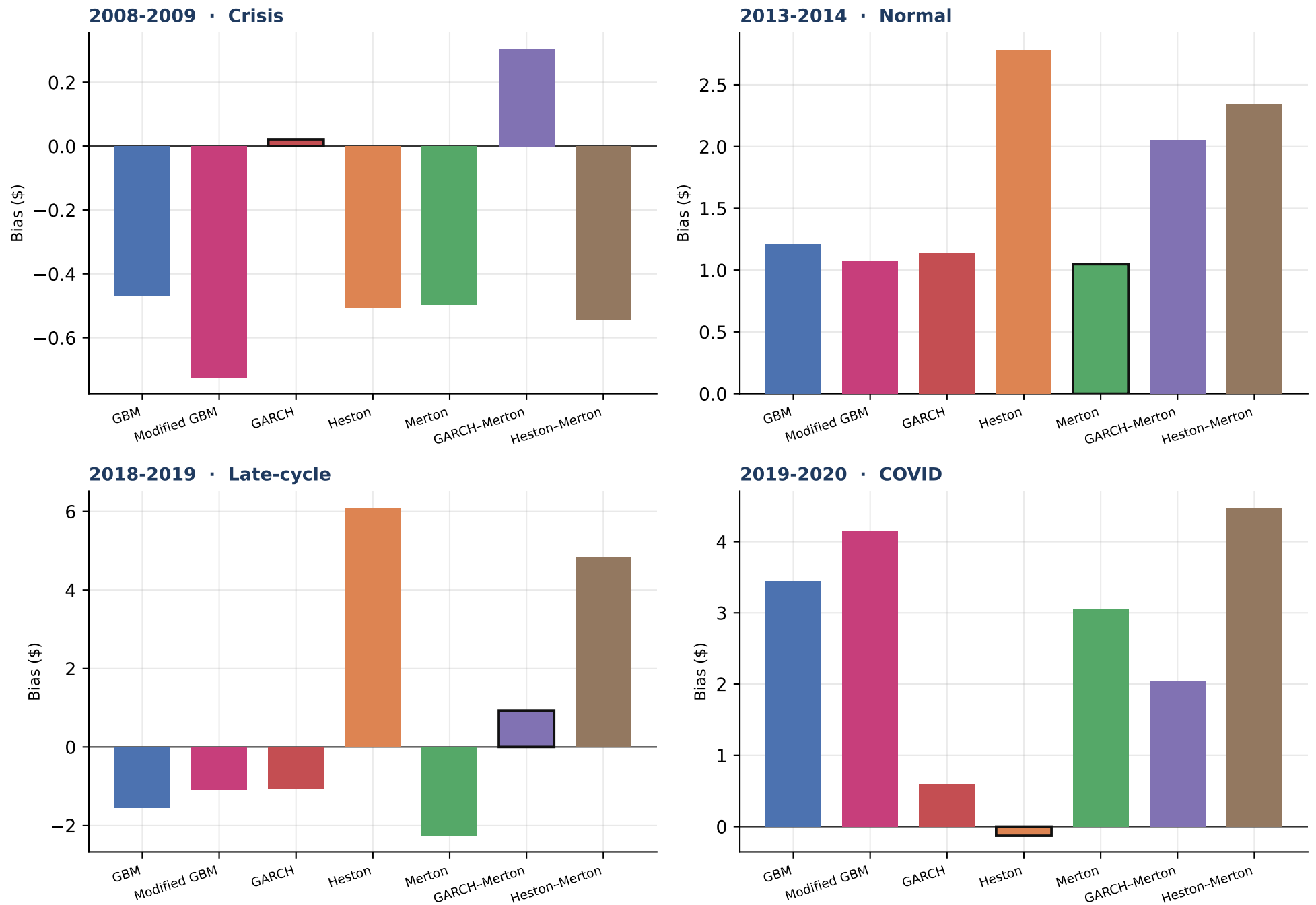
Outlined bar = lowest mean RMSE% in that regime. Equal-weight mean of the four companies.



Outlined bar = lowest mean RMSE% for that company.



Green / 1 = lowest RMSE% in that company×regime cell. Rankings can and do change across names and volatility states.



Outlined bar = closest to zero bias (not the RMSE ranking). Positive = model expensive vs listed quote.

Which models perform best, and does the ranking change?

- On mean percentage RMSE across the 16 company×regime cells, Merton is the best overall model (mean RMSE% = 36.68; 5 of 16 cells).
- Second overall is Modified GBM (mean RMSE% = 37.46; 3 cells).
- The ranking is not stable. Best-in-cell models across the 16 tables: GARCH, GARCH-Merton, GBM, Heston, Heston-Merton, Merton, Modified GBM.
- SPY: lowest mean RMSE% is Merton (39.66); wins 2 of 4 regimes.
- AAPL: lowest mean RMSE% is Merton (35.92); wins 1 of 4 regimes.
- MSFT: lowest mean RMSE% is Modified GBM (37.63); wins 2 of 4 regimes.
- AMZN: lowest mean RMSE% is GARCH (32.35); wins 0 of 4 regimes.
- 2008-2009 (Crisis): best mean RMSE% is Heston-Merton (31.98).
- 2013-2014 (Normal): best mean RMSE% is Merton (42.28).
- 2018-2019 (Late-cycle): best mean RMSE% is Modified GBM (20.89).
- 2019-2020 (COVID): best mean RMSE% is Merton (48.92).
- No-jump models as a group have lower mean RMSE% (40.14) than the jump models in this report (42.99).
- For the overall winner (Merton), crisis RMSE% is 34.18 vs 42.28 in the 2014 normal year — absolute pricing error is regime-dependent even when the ranking is not.
- These conclusions are conditional on the V3 filters, 1.5-year fixed calibration, Monday ATM sample, LSM with 10,000 paths, and American-call quotes only. They are not a statement about European options or other tenors.

How to read the 12 detailed tables

- Each of Tables SPY-1...MSFT-4 uses exactly the same option contracts for all seven models in this report. Differences are the dynamics, not the sample.
- BEST is always lowest RMSE%. MAE, Bias \$, Bias%, and early-exercise are reported, not used for the ranking.
- A model that is BEST on RMSE% can still be biased (systematically expensive or cheap).

Early-exercise fractions are a model implication, not an accuracy score; American calls on these tenors often have modest early-exercise value.

Notebook: [Reproduction of V3 Models](#) · Engine: [V3-Models_result/scripts/run_v3_1p5y_10k_fixed_empirical_study.py](#) · [payload.json](#) is the shared number store.